Solution: All Tasks Scheduling Orders

Problem Statement

There are ‘N’ tasks, labeled from ‘0’ to ‘N-1’. Each task can have some prerequisite tasks which need to be completed before it can be scheduled. Given the number of tasks and a list of prerequisite pairs, write a method to print all possible ordering of tasks meeting all prerequisites.

Example 1:

```
Input: Tasks=3, Prerequisites=[0, 1], [1, 2]
Output: [0, 1, 2]
Explanation: There is only possible ordering of the tasks.
```

Example 2:

```
Input: Tasks=4, Prerequisites=[3, 2], [3, 0], [2, 0], [2, 1]
Output:
1) [3, 2, 0, 1]
2) [3, 2, 1, 0]
Explanation: There are two possible orderings of the tasks meeting all
prerequisites.
```

Example 3:

```
Input: Tasks=6, Prerequisites=[2, 5], [0, 5], [0, 4], [1, 4], [3, 2],
[1, 3]
Output:
1) [0, 1, 4, 3, 2, 5]
2) [0, 1, 3, 4, 2, 5]
3) [0, 1, 3, 2, 4, 5]
4) [0, 1, 3, 2, 5, 4]
5) [1, 0, 3, 4, 2, 5]
6) [1, 0, 3, 2, 4, 5]
7) [1, 0, 3, 2, 5, 4]
8) [1, 0, 4, 3, 2, 5]
9) [1, 3, 0, 2, 4, 5]
10) [1, 3, 0, 2, 5, 4]
11) [1, 3, 0, 4, 2, 5]
12) [1, 3, 2, 0, 5, 4]
13) [1, 3, 2, 0, 4, 5]
```

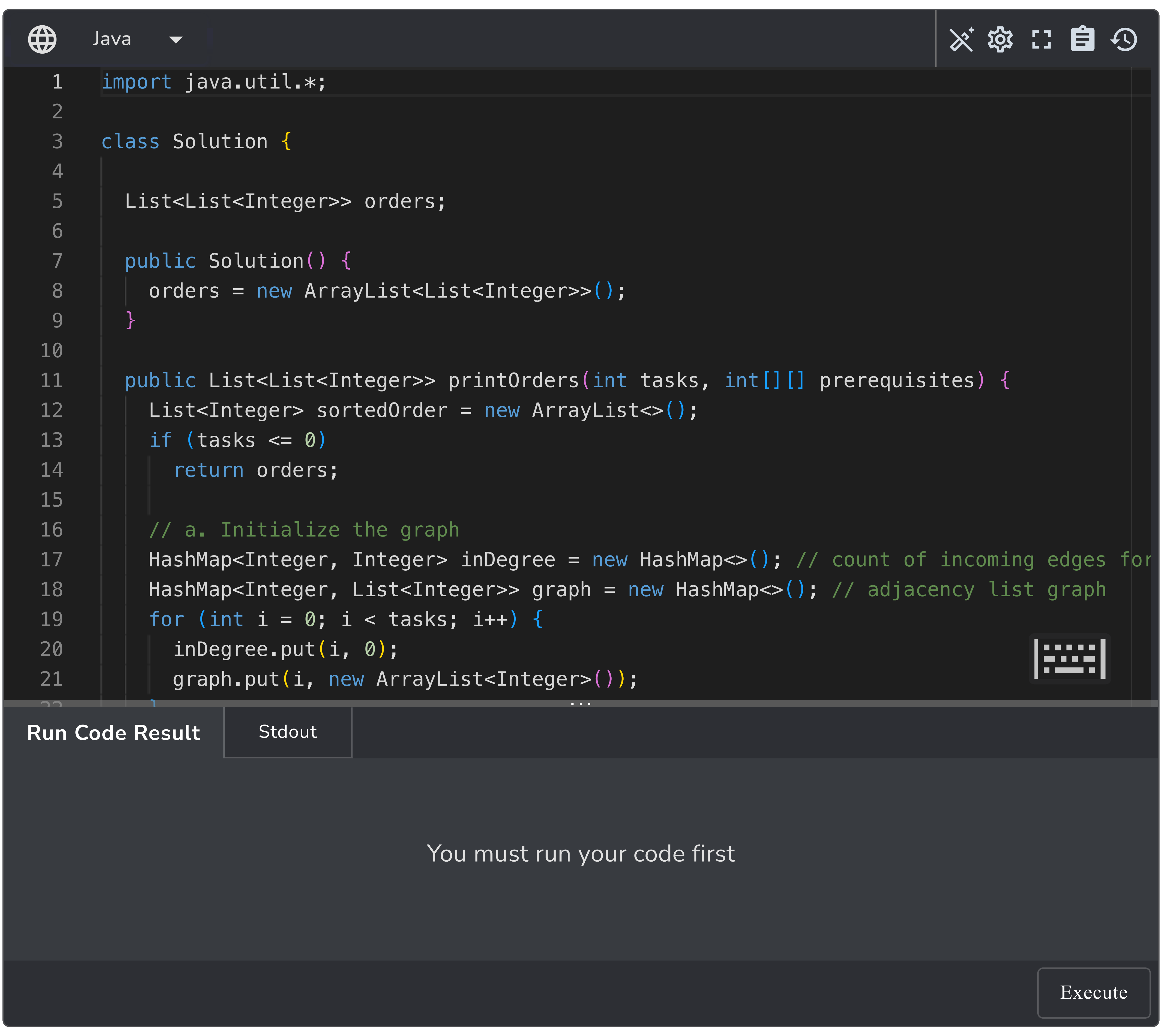
Solution

This problem is similar to `Tasks Scheduling Order`, the only difference is that we need to find all the topological orderings of the tasks.

At any stage, if we have more than one source available and since we can choose any source, therefore, in this case, we will have multiple orderings of the tasks. We can use a recursive approach with **Backtracking** to consider all sources at any step.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Solution {

  List<List<Integer>> orders;

  public Solution() {
    orders = new ArrayList<List<Integer>>();
  }

  public List<List<Integer>> printOrders(int tasks, int[][] prerequisites) {
    List<Integer> sortedOrder = new ArrayList<>();
    if (tasks <= 0)
      return orders;

    // a. Initialize the graph
    HashMap<Integer, Integer> inDegree = new HashMap<>(); // count of incoming edges for
    HashMap<Integer, List<Integer>> graph = new HashMap<>(); // adjacency list graph
    for (int i = 0; i < tasks; i++) {
      inDegree.put(i, 0);
      graph.put(i, new ArrayList<Integer>());
```

You must run your code first

Time and Space Complexity

If we don’t have any prerequisites, all combinations of the tasks can represent a topological ordering. As we know, that there can be $N!$ combinations for ‘N’ numbers, therefore the time and space complexity of our algorithm will be $O(V! * E)$ where ‘V’ is the total number of tasks and ‘E’ is the total prerequisites. We need the ‘E’ part because in each recursive call, at max, we remove (and add back) all the edges.

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